FEDERAL SPECIFICATION

ALUM, AMMONIA (ALUMINUM, AMMONIUM SULFATE), TECHNICAL

This amendment, which forms a part of Federal Specification O-A-401, dated April 10, 1953, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

Page 1, paragraph 2.1: Under the heading *"Federal Specifications:"* add the following:

U-M-186—Medicinal Products and Clinical Laboratory Reagents: General Specification for Containers (Packaging and Packing).

PPP-C-96—Cans, Metal, 28-Gage and Lighter.

PPP-B-585—Boxes; Wood Wirebound.

PPP-D-723—Drums, Fiber.

Page 2, paragraph 3.1: Delete table and substitute the following:

TABLE I.—*Chemical requirements*

	Percent by weight
Alumina (Al_2O_3) ----------minimum--	11.0
Ammonia (NH_3) ----------minimum--	3.50
Sulfur trioxide (SO_3) ------minimum--	35.00
Water insoluble matter ----maximum--	0.5

Page 4, paragraph 4.3.5: Delete in its entirety and substitute the following:

4.3.5 *Sulfur trioxide content.*—Take a 100-ml. aliquot of the prepared solution (see 4.3.1) and place in a 400-ml. beaker. Make the solution acid by the addition of hydrochloric acid (sp. gr. 1.2) using phenolphtalein as the indicator, and add 4 ml. of hydrochloric acid in excess. Dilute the solution to approximately 250 ml. with hot water and heat to boiling. Add 20 ml. of a 10-percent solution of barium chloride from a burette at a rate of 10-ml. in 2 to 10 minutes. Place the beaker on a steam bath and digest for an hour. Filter the solution through a tared Gooch crucible. Pass the filtrate through the same filter a second time, and wash the precipitate 10 times with hot water. Dry the precipitate and place in a muffle furnace for half an hour at 800°C. and weigh.

$$\text{Percentage sulfur trioxide } (SO_3) = \frac{34.3A}{B}$$

where:

A = weight of barium sulfate.

B = weight of sample in aliquot.

Page 4, section 5, delete in its entirety and substitute:

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A, military package.*

5.1.1.1 *Capsules (class 2).*—Eighty capsules of the material shall be packaged in an eight-ounce capacity wide mouth screw cap type glass bottle conforming to the requirements of Federal Specification U-M-186, type I, class A. The cap shall be held in place with a self-shrinking plastic film.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

5.1.1.2 *Nut size (class 1).*—Ten pounds of nut size ammonia alum shall be packaged in a ten-pound capacity, painted, round metal can with a multiple friction plug closure. The can shall conform to the requirements of Federal Specification PPP–C–96, type V, class 2, finish plan B.

5.1.2 *Level C, minimum military and civilian package.*

5.1.2.1 *Capsules (class 2).* — Same as 5.1.1.1.

5.1.2.2 *Nut size (class 1).*—Ten pounds of nut size ammonia alum shall be packaged in a ten-pound capacity multiple friction type round metal can of substantial commercial grade commonly used for the purpose.

5.2 Packing.

5.2.1 *Level A, military pack.*

5.2.1.1 *Capsules (class 2).* — Forty-eight eight-ounce bottles of capsules shall be packed in a snug fitting "V" board box conforming to the requirements of Military Specification JAN–P–108, style RSC. The arrangement within this box shall be two tiers floored 4 by 6. Snug fitting, full height, half-slotted style fiberboard partitions shall be used. Full size liners, top, bottom and layer pads shall also be used. Partitions, liners and pads shall be fabricated from W5c (JAN–P–108) fiberboard. Closure of this box shall be effected as specified in Appendix I of Military Specification JAN–P–108.

5.2.1.2 *Nut size (class 1).*

5.2.1.2.1 *For indeterminate use.*—Six ten-pound cans shall be packed in a snug fitting "V" board box conforming to the requirements of Military Specification JAN–P–108, style RSC. Arrangement within this box shall be one tier floored 2 by 3. Closure of the box shall be effected as specified in Appendix I of Military Specification JAN–P–108. The fiberboard box shall then be overpacked in a wirebound Q-crate so designed as to prevent the box from falling out the open ends. Structure and closure of this crate shall be the same as specified in Federal Specification PPP–B–585, class 3 use, except that ends are not required.

5.2.1.2.2 *For normal supply*[1].—Same as set forth in 5.2.1.2.1, except that use of the Q–crate is not required.

5.2.1.3 *One hundred pound quantities (class 1).*—One hundred pounds of class 1 material shall be packed in a one-hundred-pound-capacity fiber drum conforming to the requirements of Federal Specification PPP–D–723, type III, grade A. In addition the the drum shall be provided with a water-vaporproof barrier, lining or interior coating.

5.2.2 *Level B, limited military pack.*

5.2.2.1 *Capsules (class 2).* — Forty-eight eight-ounce bottles of capsules shall be packed in a snug fitting fiberboard box conforming to the requirements of Federal Specification LLL–B–631, style RSC. The arrangement within this box shall be two tiers floored 4 by 6. Snug fitting, full height, half-slotted style fiberboard partitions shall be used. Full size liners and top, bottom and layer pads shall also be used. Partitions, liners and pads shall be fabricated of the same material as used in the box. The box shall be sealed in such a manner that its closure will pass the closure test outlined in the box specification.

5.2.2.2 *Nut size (class 1).*

5.2.2.2.1 *Ten-pound quantities.* — Six ten-pound cans of nut size ammonia alum shall be packed in a snug fitting fiberboard box, conforming to the requirements of Federal Specification LLL–B–631, style RSC. The arrangement within this box shall be one tier floored 2 by 3. The box shall be sealed in

[1] Shipment in established channels to known destinations, and storage under favorable conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.

such a manner that its closure will pass the closure test outlined in the applicable specification.

5.2.2.2.2 *One hundred pound quantities.*— One hundred pounds of nut size ammonia alum shall be packed in a one hundred pound capacity fiber drum conforming to the requirements of Federal Specification PPP–D–723, type I, grade A. In addition the drum shall be provided with a water-vaporproof barrier, lining or interior coating.

5.2.3 *Level C, minimum military and civilian pack.*

5.2.3.1 *Capsules (class 2).* — Forty-eight eight-ounce bottles of capsules shall be packed in a corrugated fiberboard box conforming to the requirements of Rule 41 of the Uniform Freight Classification. Arrangement within this box shall be two tiers floored 4 by 6.

5.2.3.2 *Nut size (class 1).*

5.2.3.2.1 *Ten-pound quantities.* — Six ten-pound cans of nut size ammonia alum shall be packed in a fiberboard box conforming to the requirements of Rule 41 of the Uniform Freight Classification. Arrangement within this box shall be one tier floored 2 by 3.

5.2.3.2.2 *One hundred pound quantities.*— One hundred pounds of nut size ammonia alum shall be packed in a one hundred pound capacity fiber drum conforming to the requirements of Rule 41 of the Uniform Freight Classification. In addition the drum shall be provided with a water-vaporproof barrier, lining or interior coating. When authorized in the invitation to bid, one hundred pounds shall be packed in a 4-ply sewn open-mouth paper shipping sack with plies of 1/40, 2/50, and 1/60 and conforming to the requirements of Rule 40 of the Uniform Freight Classification.

5.3 **Marking.**—In addition to any special marking required below or in the contract or order, interior packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL–STD–129.

5.3.1 *Special marking.*—Each bottle of capsules shall be labeled as follows:

ALUM, AMMONIA
This material is Ammonia Alum, 1 part Ammonia Alum equals 2/3 part Aluminum Sulfate (filter alum).

MILITARY INTERESTS:
Army—C E
Navy—Y S.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:13Z
Check the source to verify that this is the current version before use.